GRADING STATS REPORT 3.0

DATA ANALYTICS AND VISUALIZATION TEAM

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Topics Covered

A sequential overview of our analysis

1. Average Grades of Freshies
2. Average Grades of Sophies
3. Average Grades of Thirdies
4. Department wise Electives Average Grades
5. Department wise Minors Average Grades
6. CBS Analysis & Top 30 Courses
7. Department wise Average Grades
8. Department wise Autumn & Spring Average Grades
9. AP Grade Distribution
10. Number of Courses Vs Percentage of AAs
11. Number of Courses Vs Average Grades
12. Course Recommender

*Click on the topic name to navigate to it's slide directly

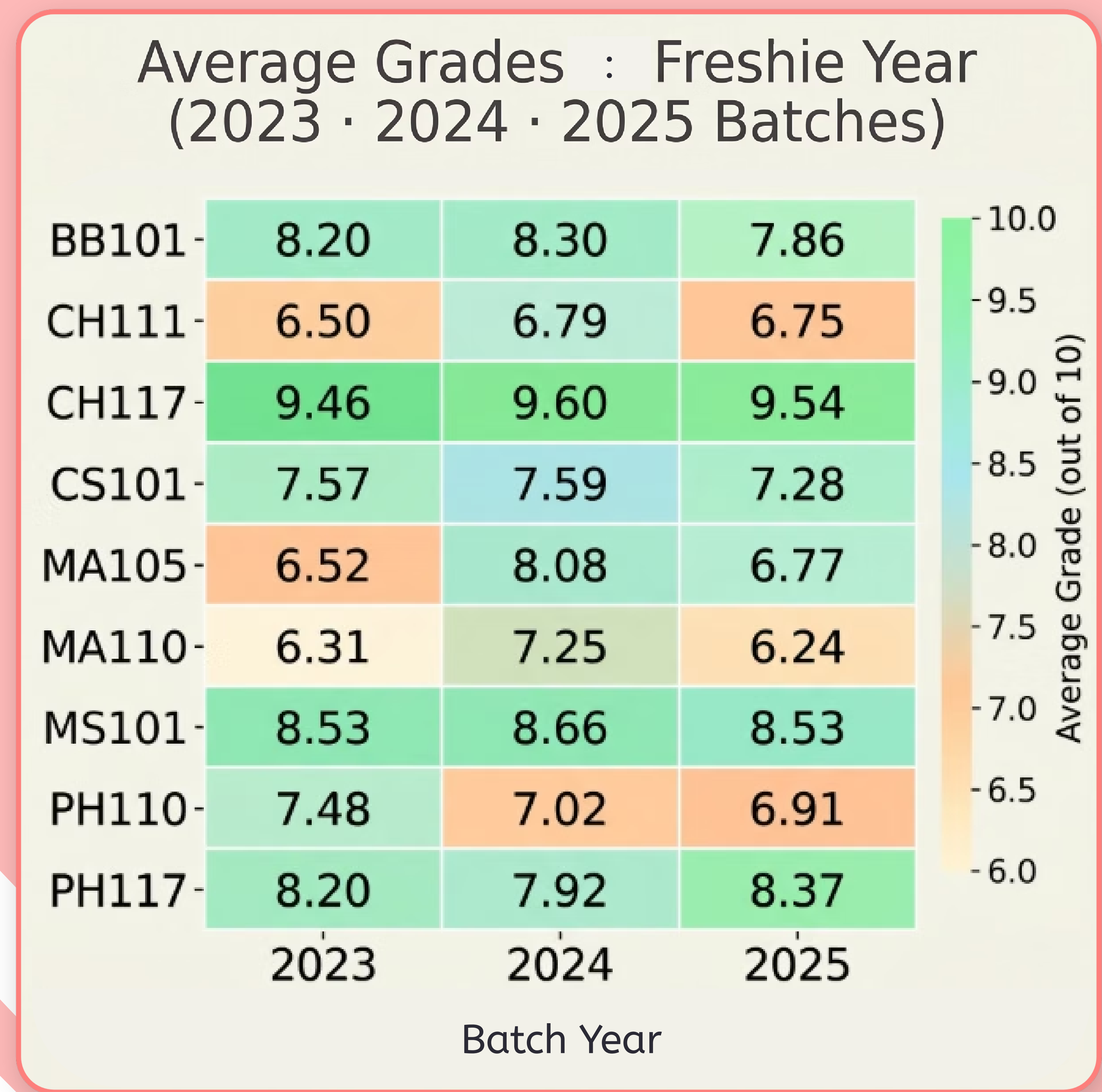
- We have used **Internal ASC** for collecting data and creating all the databases used in this report.
- We have used only the courses that are open to the **Undergraduate** students for all our analyses.

Formula used for the average grade calculations:

$$\text{CourseAverage} = \frac{\sum(\text{GradePoint} \times \text{StudentCount})}{\text{TotalStudents (excluding NP | PP)}}$$

$$\text{Dept. Average} = \frac{\Sigma(\text{CourseAverage} \times \text{Credits})}{\Sigma(\text{Credits})}$$

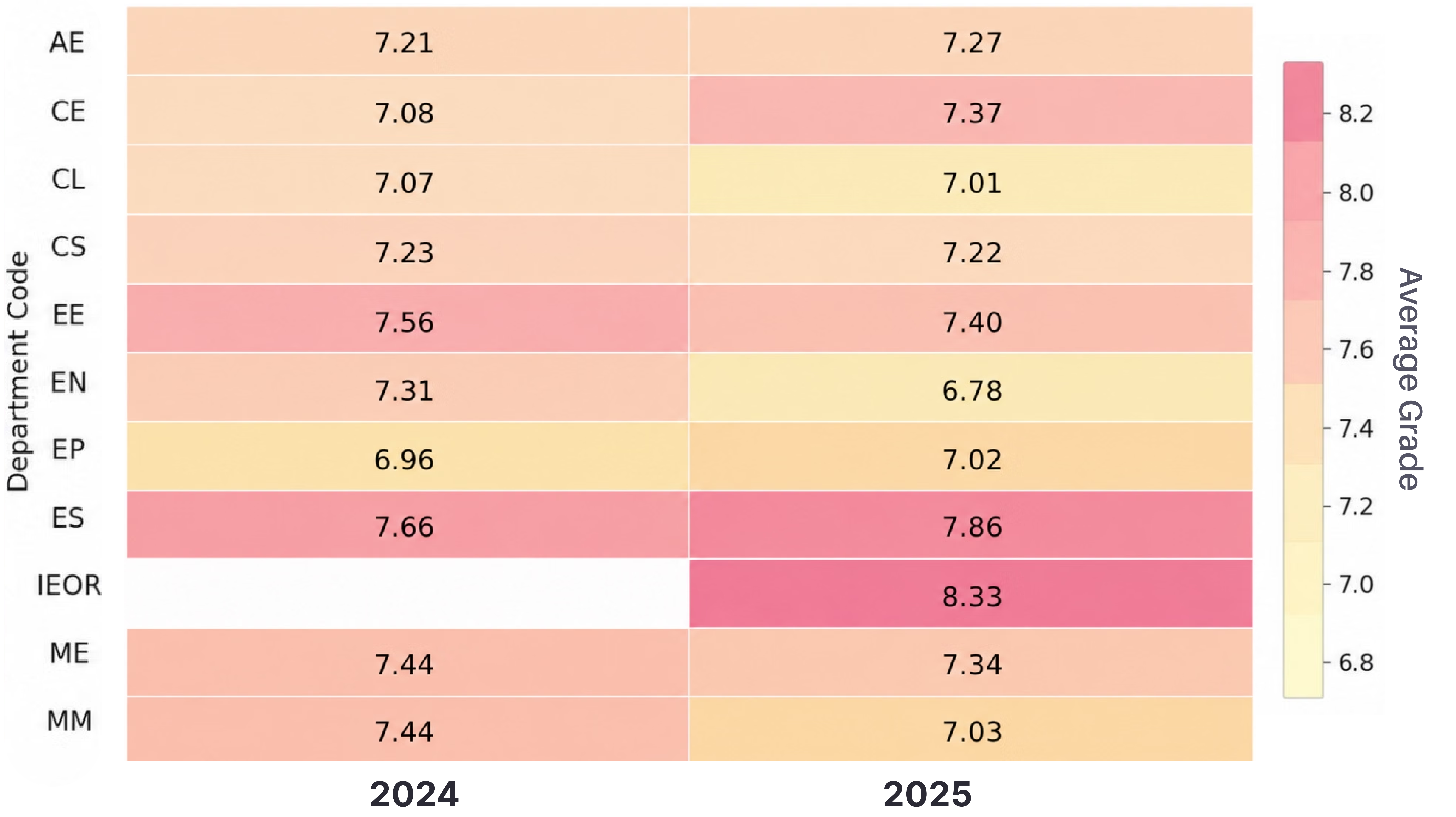
First Year Average Grades of Common Courses



The 2024 batch experienced significant grade inflation across several high credit freshman courses (MA105, MA110, MS101, and CS101). Averages peaked across the board, driven heavily by the mathematics department, where MA105 increased by 1.56 points and MA110 climbed by nearly a full point compared to 2023.

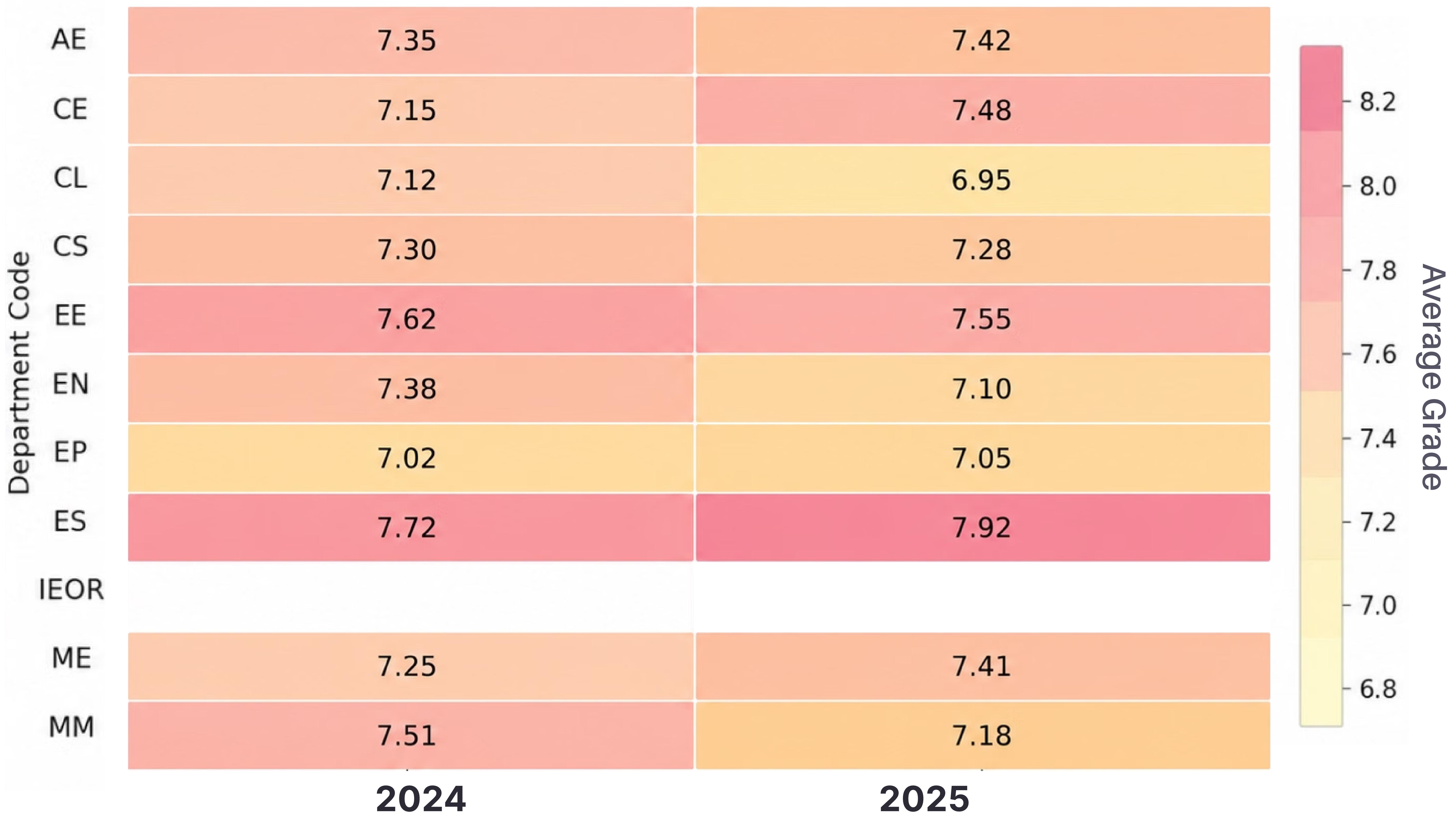
In contrast, the 2025 batch saw a distinct downward shift. Averages dropped across all major core courses: MA105 fell by 1.31 points, MA110 reverted to its 2023 baseline, BB101 hit a three year low, and both CS101 and MS101 registered notable declines.

Mandatory Core Courses Average Grades for Sophomores



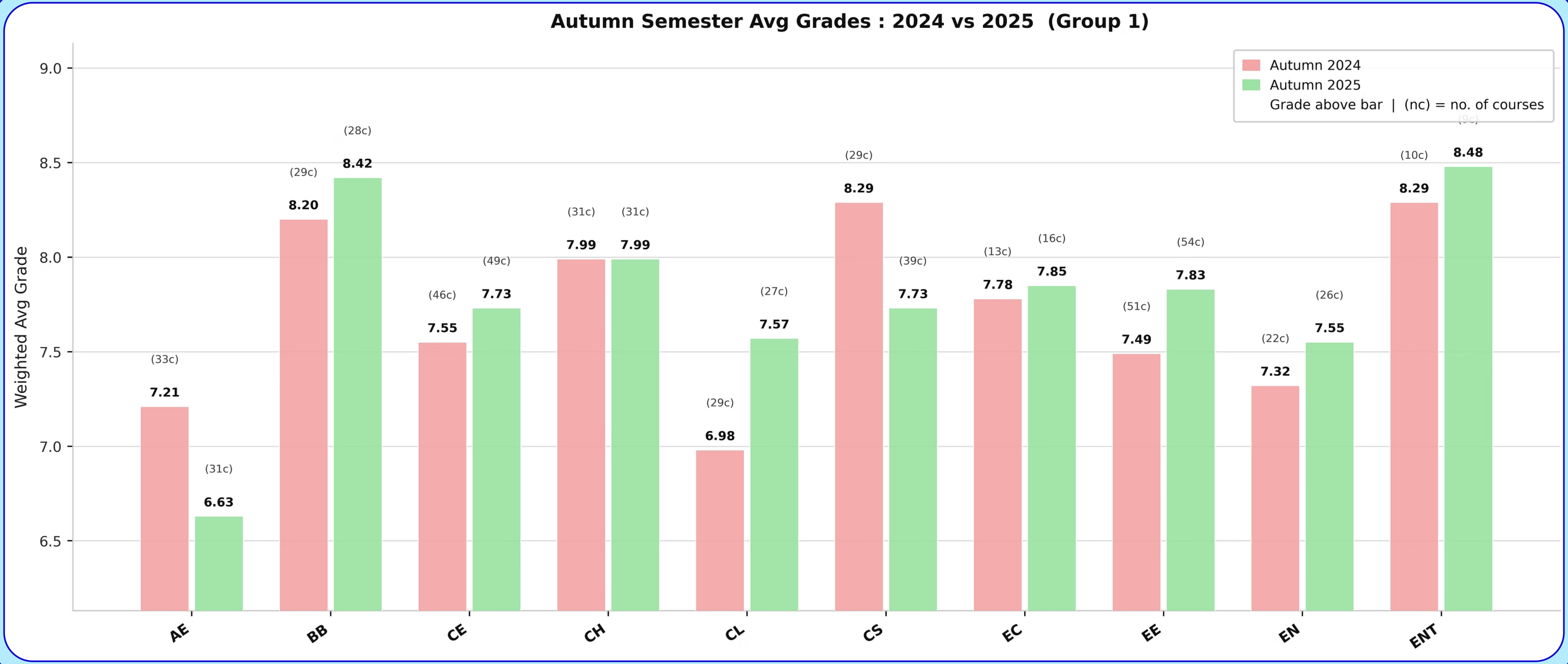
This heatmap compares the average grades of second year students across the 2024 and 2025 academic batches. The average grades are calculated exclusively using mandatory department core courses only.

Mandatory Core Courses Average Grades for Thirdies

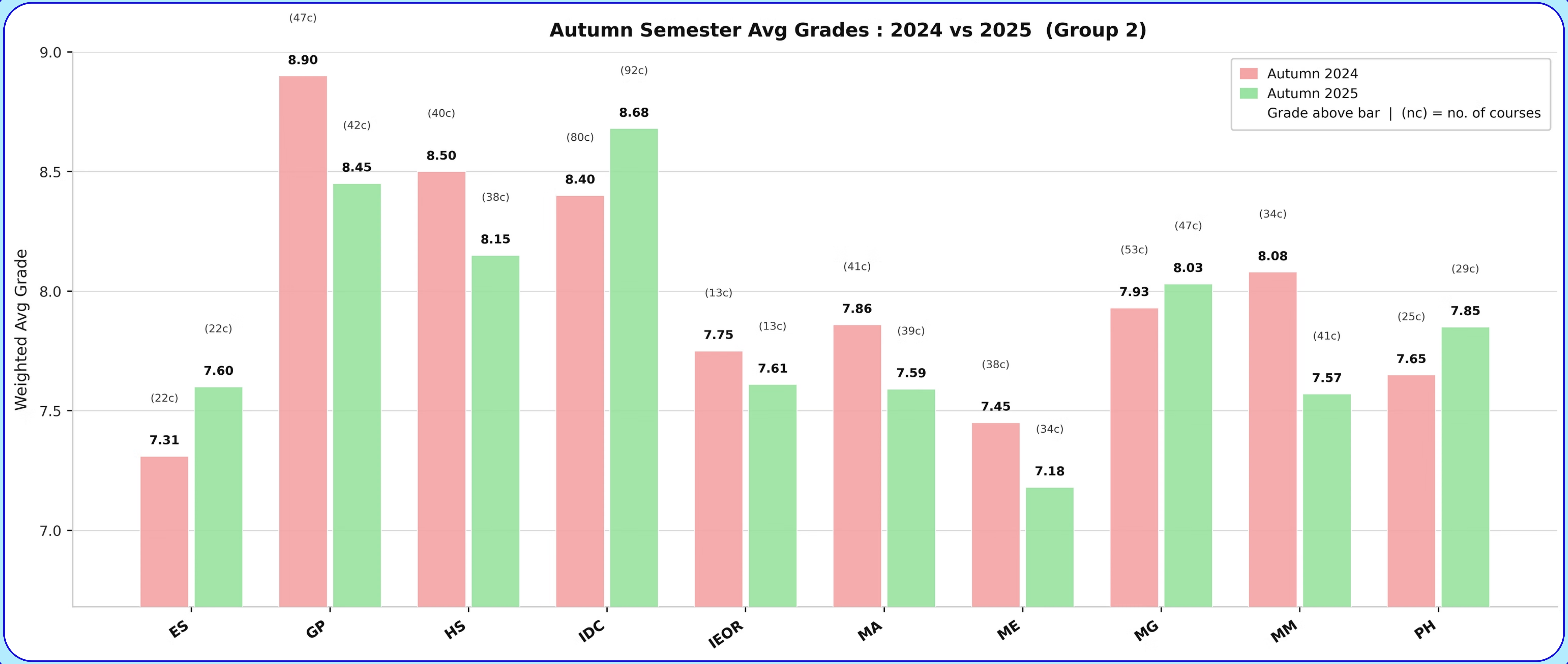


This heatmap compares the average grades of third year students across the 2024 and 2025 academic batches. The average grades are calculated exclusively using mandatory department core courses only.

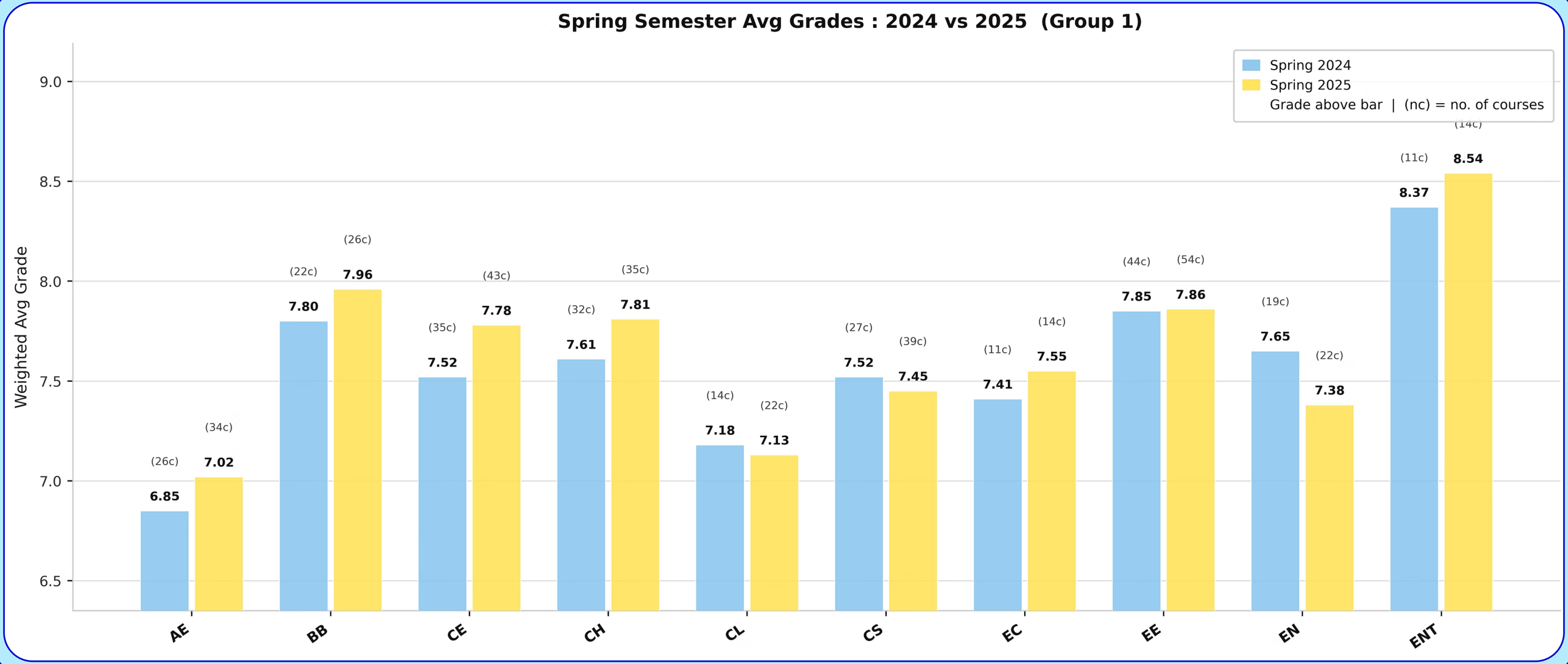
Department Wise Electives Average Grades (Autumn)



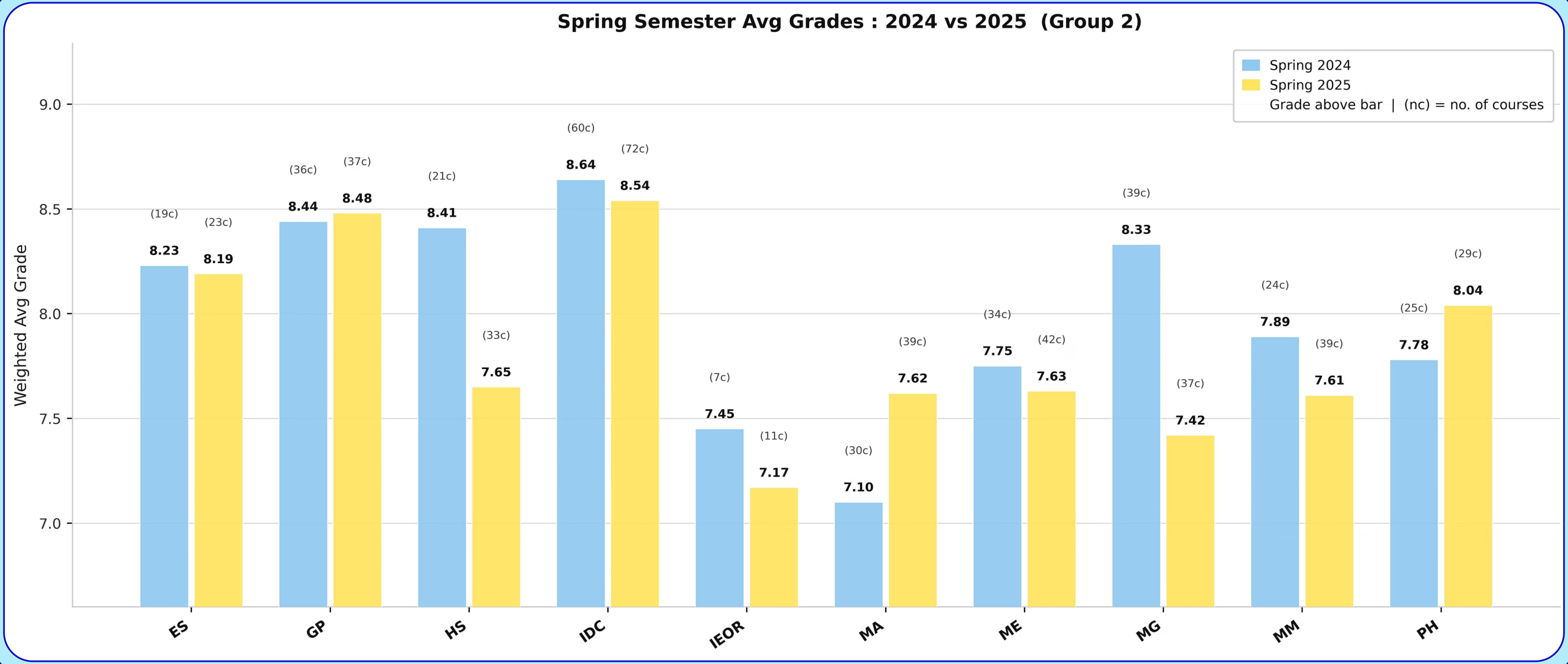
Department Wise Electives Average Grades (Autumn)



Department Wise Electives Average Grades (Spring)



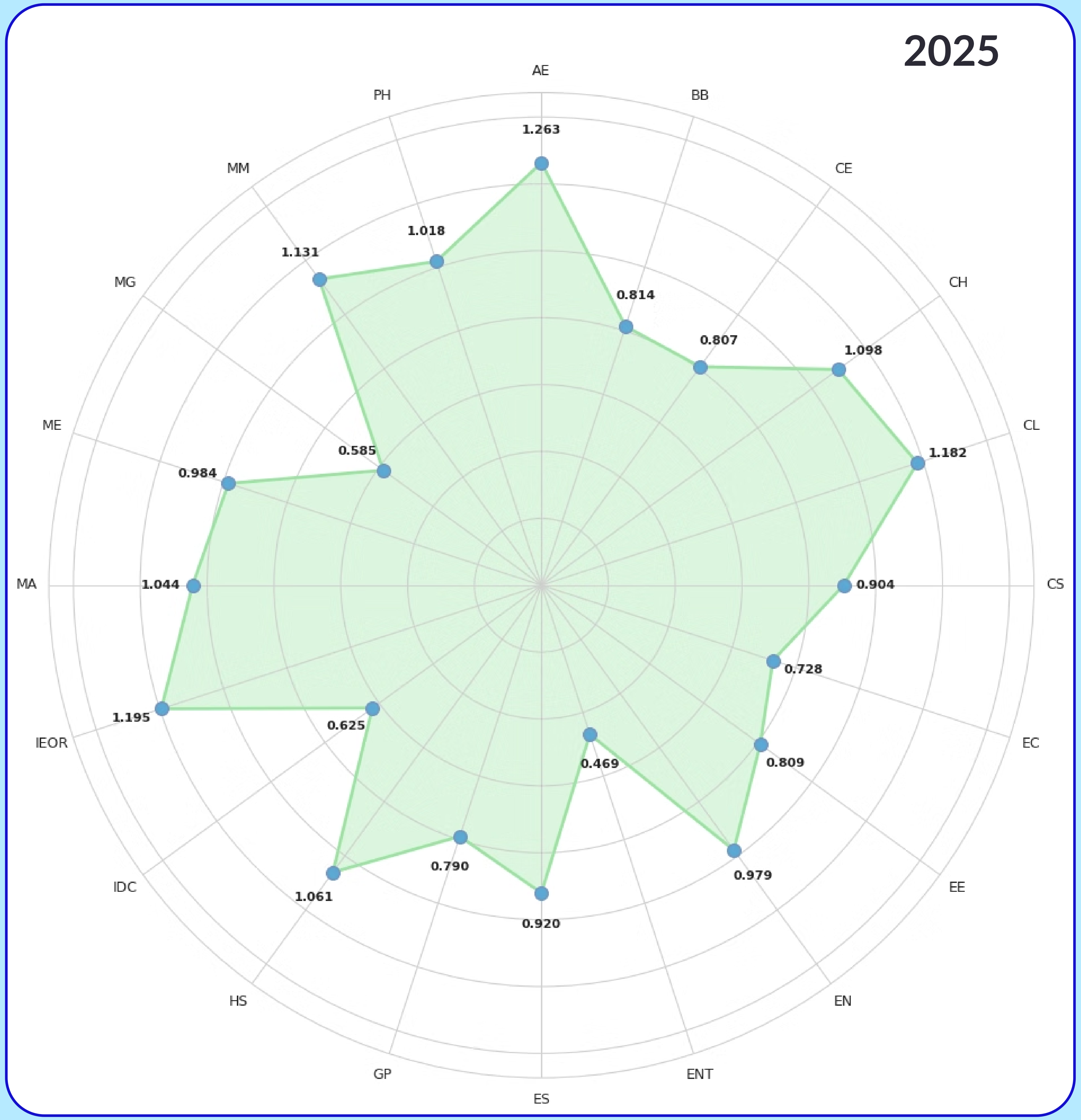
Department Wise Electives Average Grades (Spring)

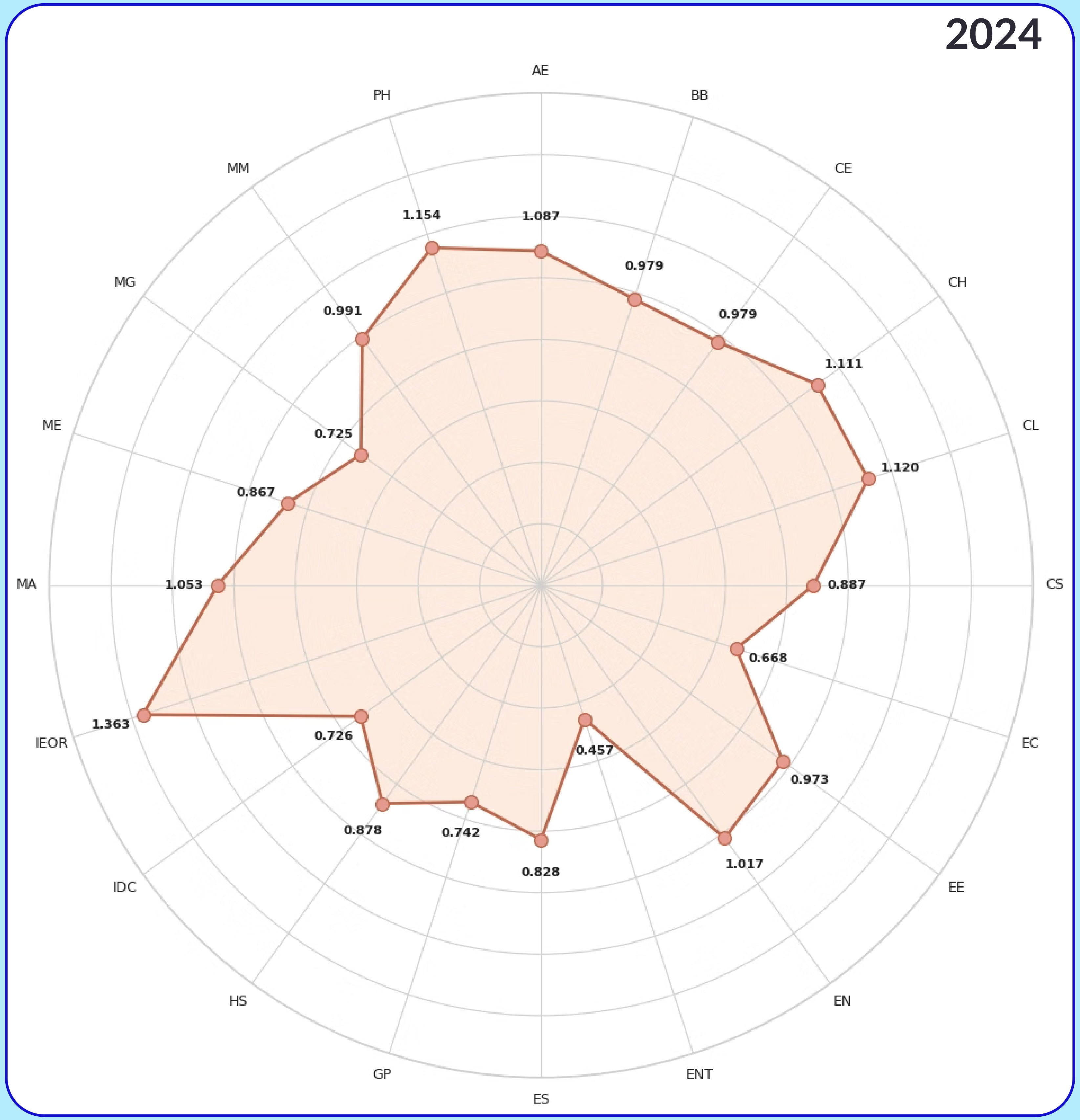


Department Wise Electives Average Grades Analysis

- Departments such as Earth Sciences, IDC Design, Entrepreneurship, Humanities & Social Sciences, Management and Biosciences & Bioengineering consistently represent the highest scoring bracket, maintaining strong elective averages between 8.0 and 8.6.
- Average grades in electives of Chemistry, Physics, Metallurgical Engineering & Materials Science, Environmental Science, Electrical Engineering, Computer Science & Engineering, Economics and Civil Engineering cluster between 7.5 and 8.0.
- Average grades in electives of Mathematics/ASI, Energy Science, Mechanical Engineering, IEOR, Aerospace Engineering and Chemical Engineering cluster between 7.1 and 7.5.
- Looking at specific courses reveals how drastically grading patterns can fluctuate year over year, even with similar class sizes. For instance, SOM610 (Marketing Management II) recorded a notably low average of 5.65 with 130 students in Autumn 2024, but increased by a massive two full points to 7.65 with 118 students in Autumn 2025.

Department Wise Electives Standard Deviation of Average Grades





Why Standard Deviation ?

Formula used :

$$SD = \sqrt{\frac{\sum (x_i - \bar{x})^2}{N}}$$

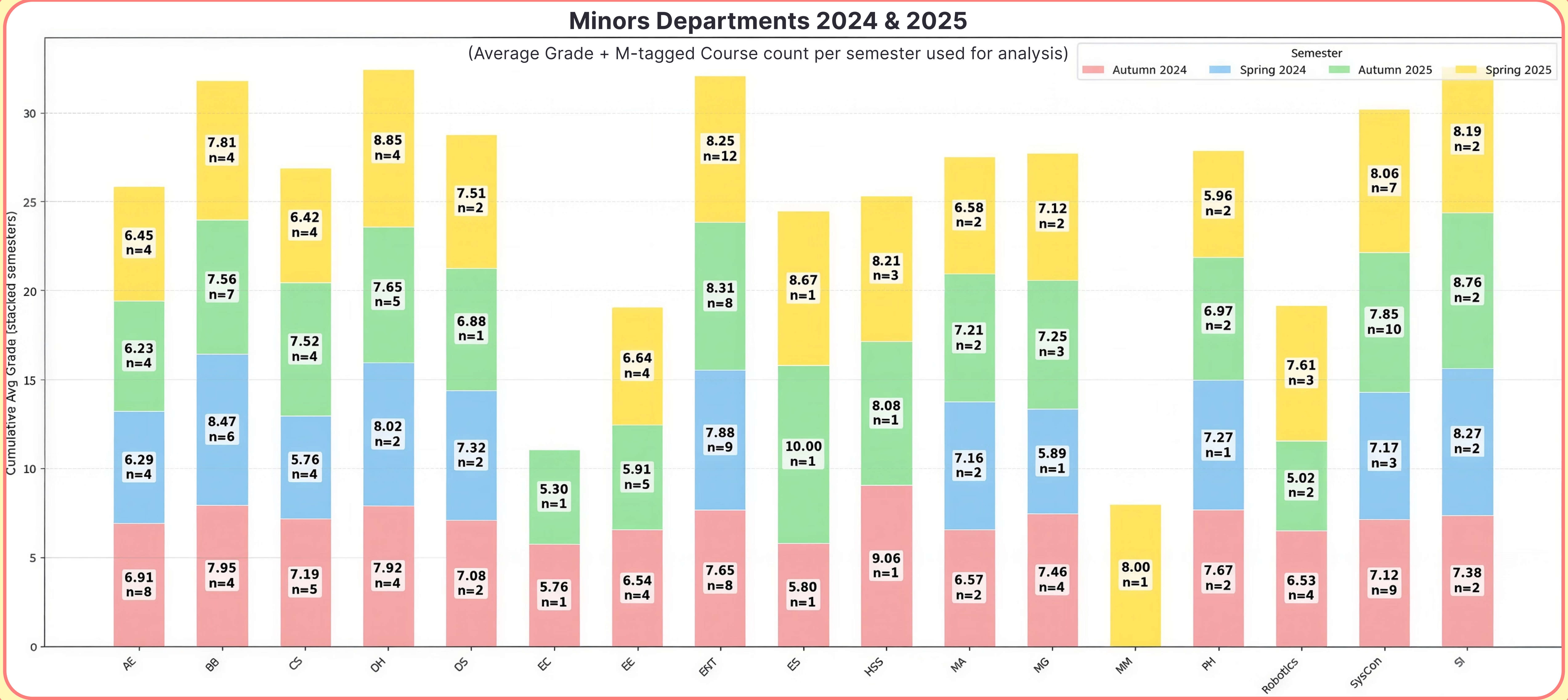
- x_i = each course's (weighted) average
- $\bar{x}$ = mean of those averages
- N = number of courses in the group

Standard deviation measures how spread out students' grades are. For example, if two courses have the same average grade (e.g., 7.5), a course with a higher standard deviation will have grades more widely dispersed. This means higher chances of both top grades (10) and low grades (6) : a "high risk, high reward" scenario. In contrast, a course with lower standard deviation will have grades more concentrated around 7 or 8, offering a "safer route" with fewer extremes.

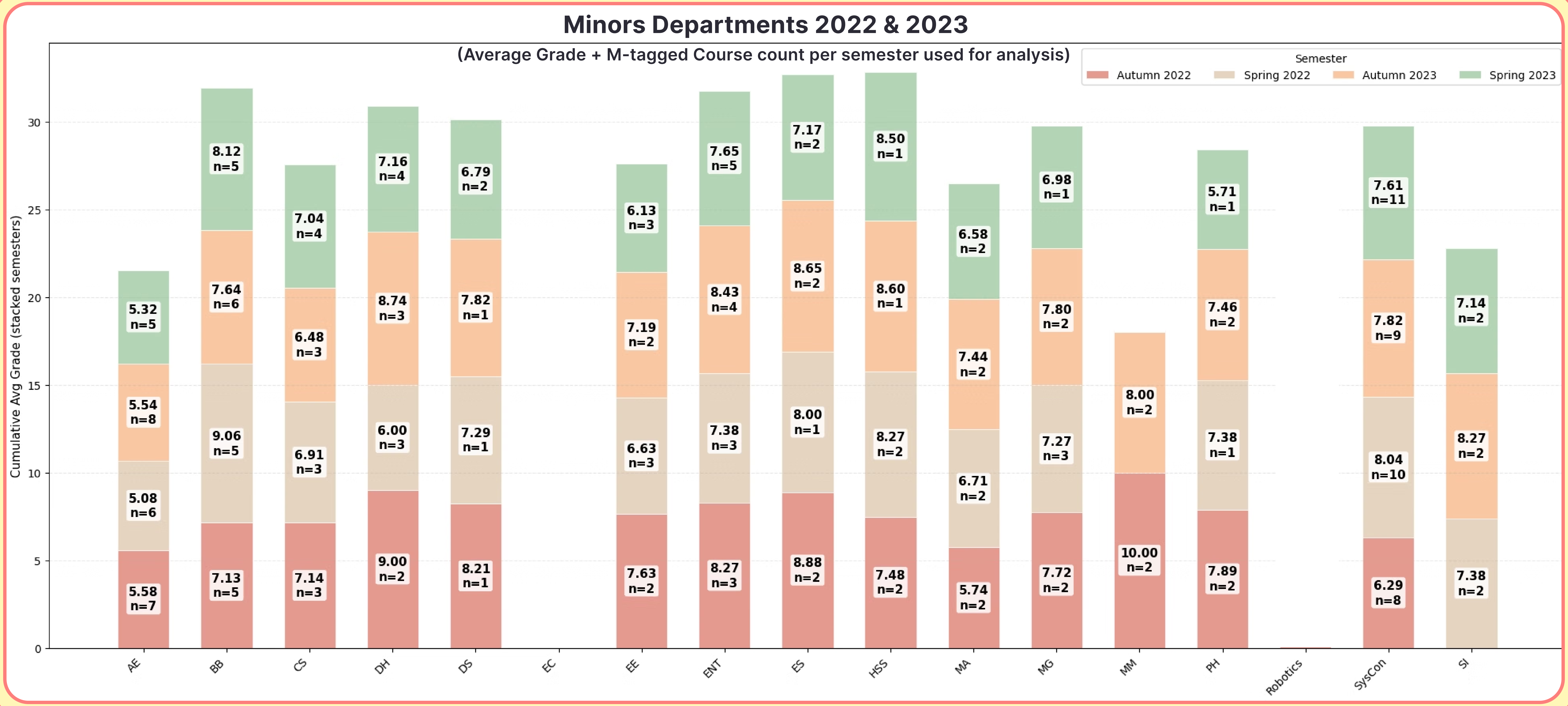
How we calculated the standard deviation?

To find a department's overall standard deviation, we first determine the weighted average grade for each of its individual course offerings. We then measure the statistical spread of these distinct course averages relative to the department's collective mean.

Department Wise Minors Average Grades



Department Wise Minors Average Grades

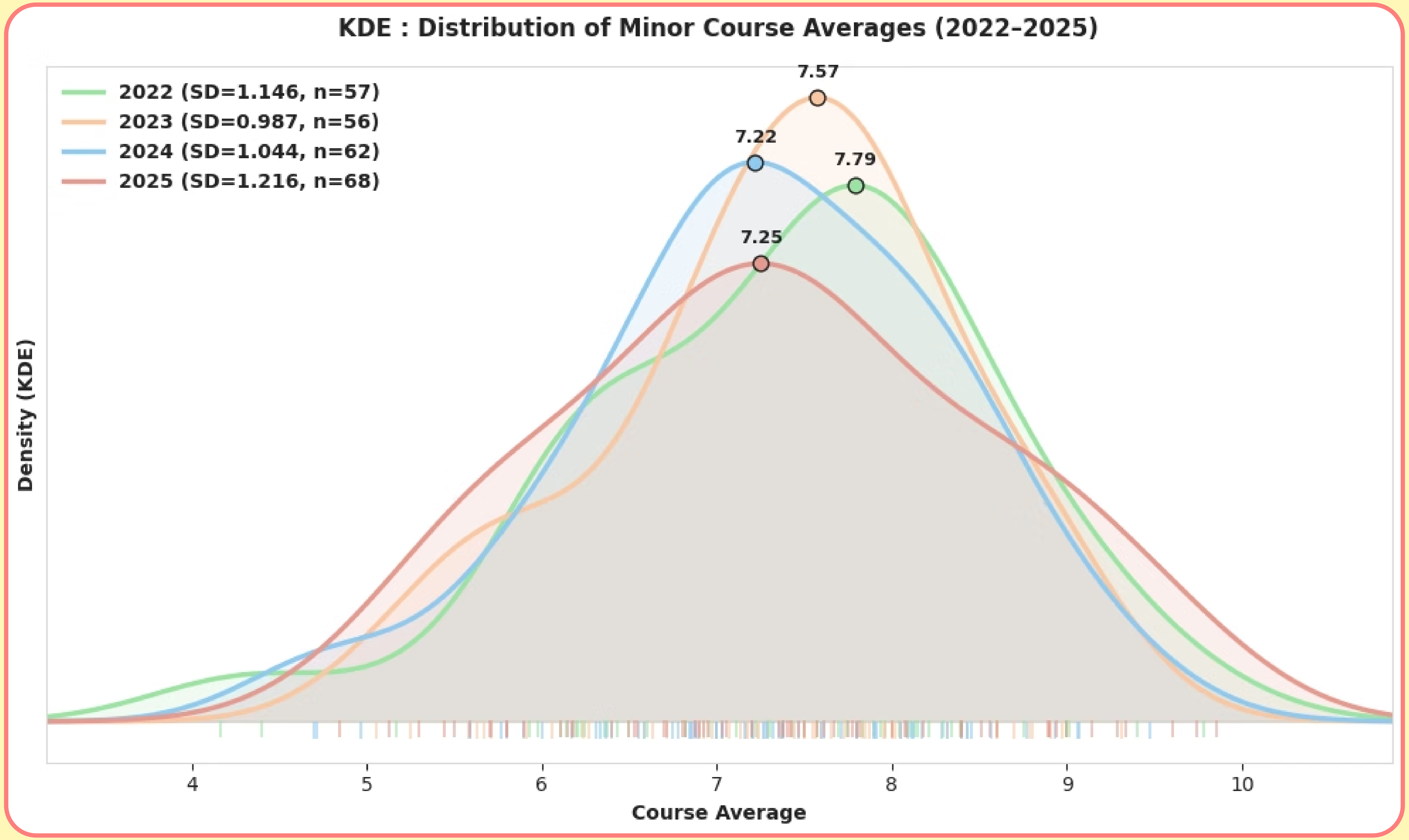


Department Wise Minors Average Grades

- **Scope:** Covers the 17 minor departments running in the upcoming 2026–27 academic year (16 home departments + Robotics), using data tracked from 2022 to 2025.
- **Course Standardization:** Focuses exclusively on core M tagged courses under the Division column to establish a consistent departmental baseline. Fluctuation heavy minor electives are excluded, as they vary entirely by individual student choice.
- **Robotics Overlap:** Robotics courses are cross listed from courses of Home departments . For example, *EE 309 (Microprocessors)* counts identically for both Electrical and Robotics since it is M-tagged in both.

Note on Data Reliability: Keep in mind that some departments have much more data than others. While Entrepreneurship (37 courses), Systems & Control (29), and BSBE (21) have solid baselines, others like MM (1), Earth Sciences (2), Energy (3), and Humanities (5) have very few data points. Averages for these low sample departments should be taken as rough indicators rather than final proof.

Standard Deviation of Year Wise Minors Average Grades



Distribution Stability vs. Widening Variance: Peak grade concentration(mode) stays steady around 7.22–7.79 across all years, but course average spread widens over time, from the tightest distribution in 2023 (SD=0.987) to the flattest in 2025 (SD=1.216).

CBS Analysis

Eager to know about the CPI boosting courses?

Voila! We got you covered!

THE CPI BOOSTER SCORE (CBS)

Credits: Rudra G

CBS is defined as : -

$$CBS = \sum_{\text{semesters in analysis}} \frac{(\text{weight}) \times F_i}{(\text{NS})}$$

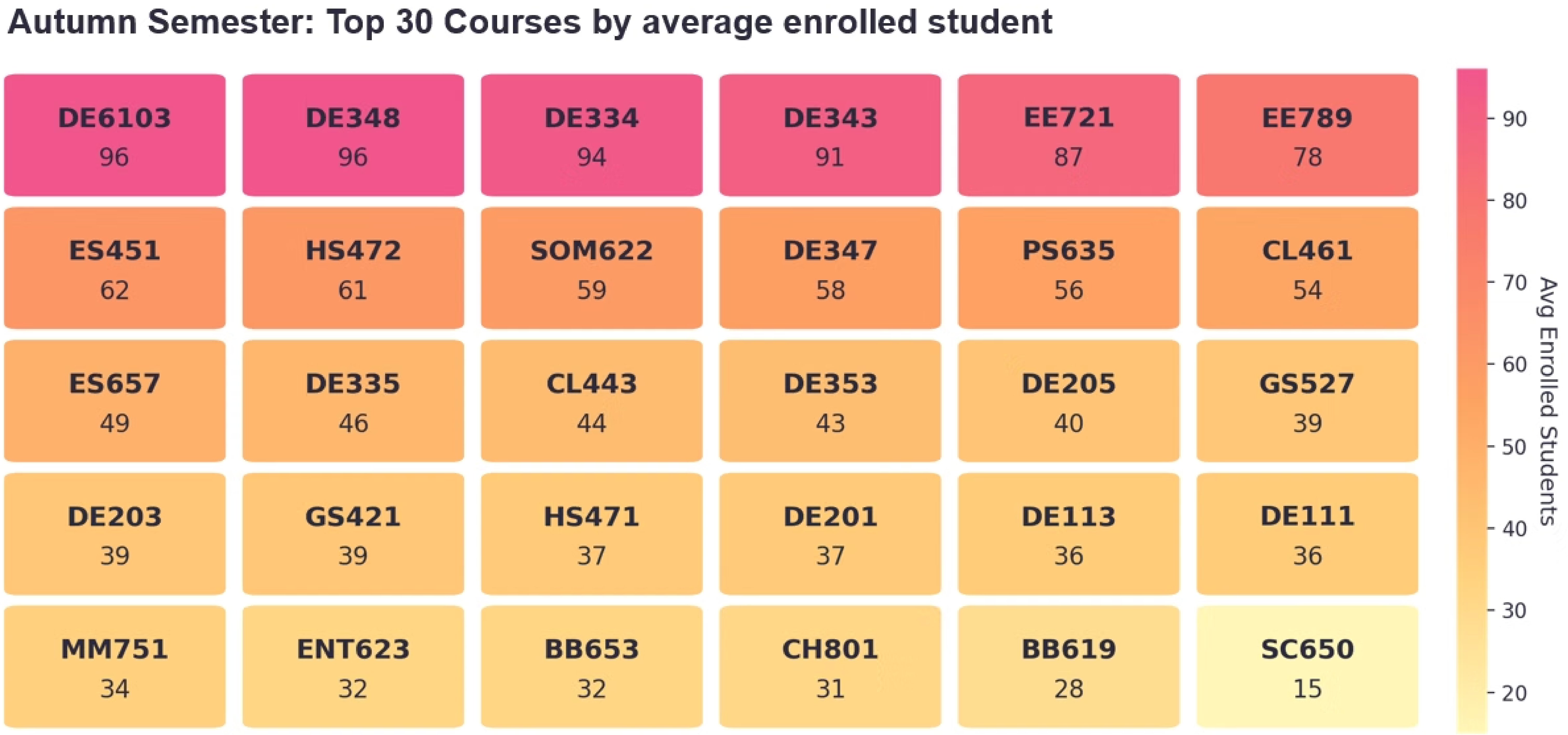
F is defined as follows : -

$$F(X, N, \sigma^2, N_0, k) = \frac{X}{\left(1+(10^{-\frac{N}{N_0}})\sigma^2\right)\left(1+\left(\frac{N_0^k}{N_0^k+N_k}\right)L\right)}$$

CBS is a score defined to assess which courses are most effective for improving CPI. A higher CBS indicates that a course is better suited for enhancing CPI.

For more details about the model, variable definitions as well as the methodology, please refer to the documentation: {link}

Top 30 Courses Based On CBS (Autumn)



*We neither endorse nor discourage any courses; this analysis is solely for statistical purposes.

Top 30 Courses Based On CBS (Spring)

Spring Semester: Top 30 Courses by average enrolled student

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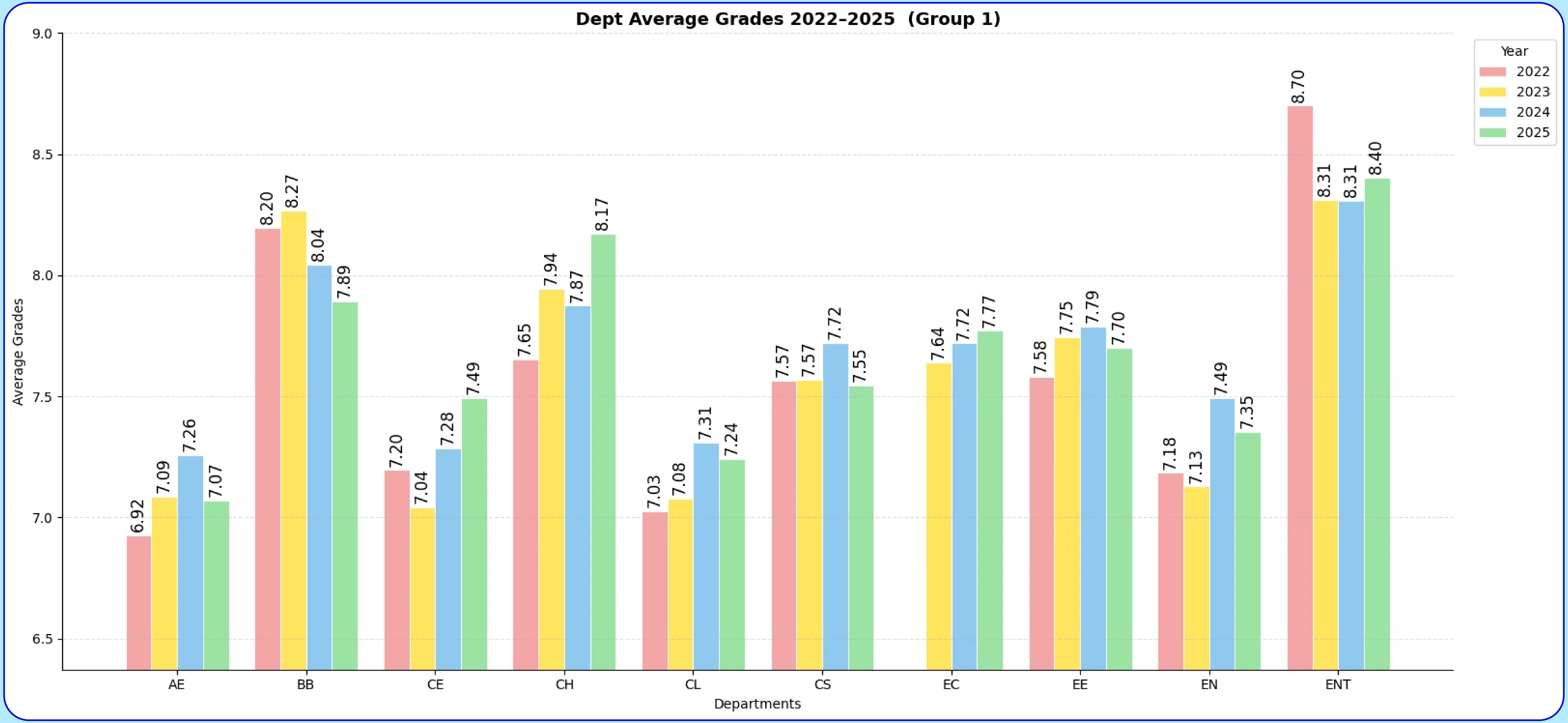
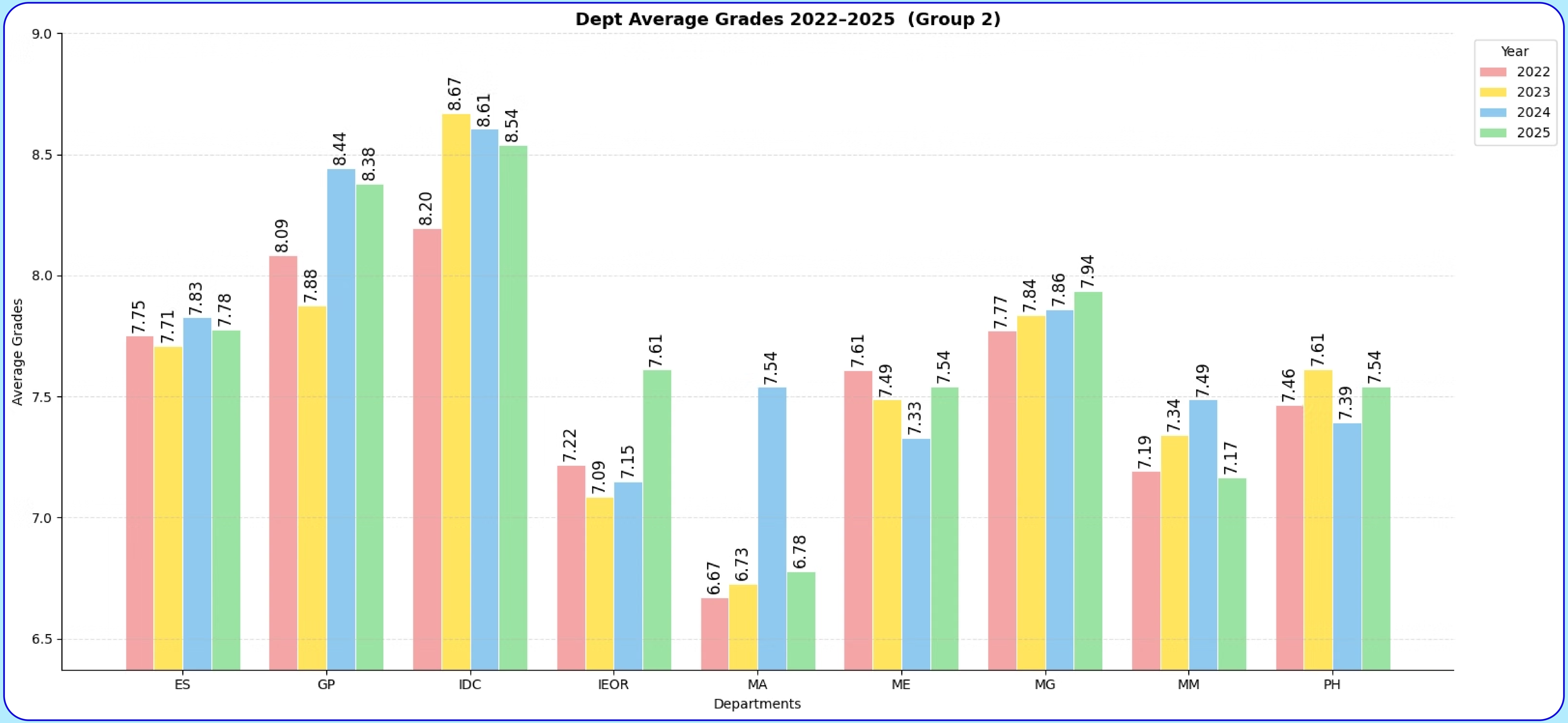
Overall Department wise Average Grades Across Years



Overall Department wise Average Grades Across Years



Overall Department wise Average Grades Across Years

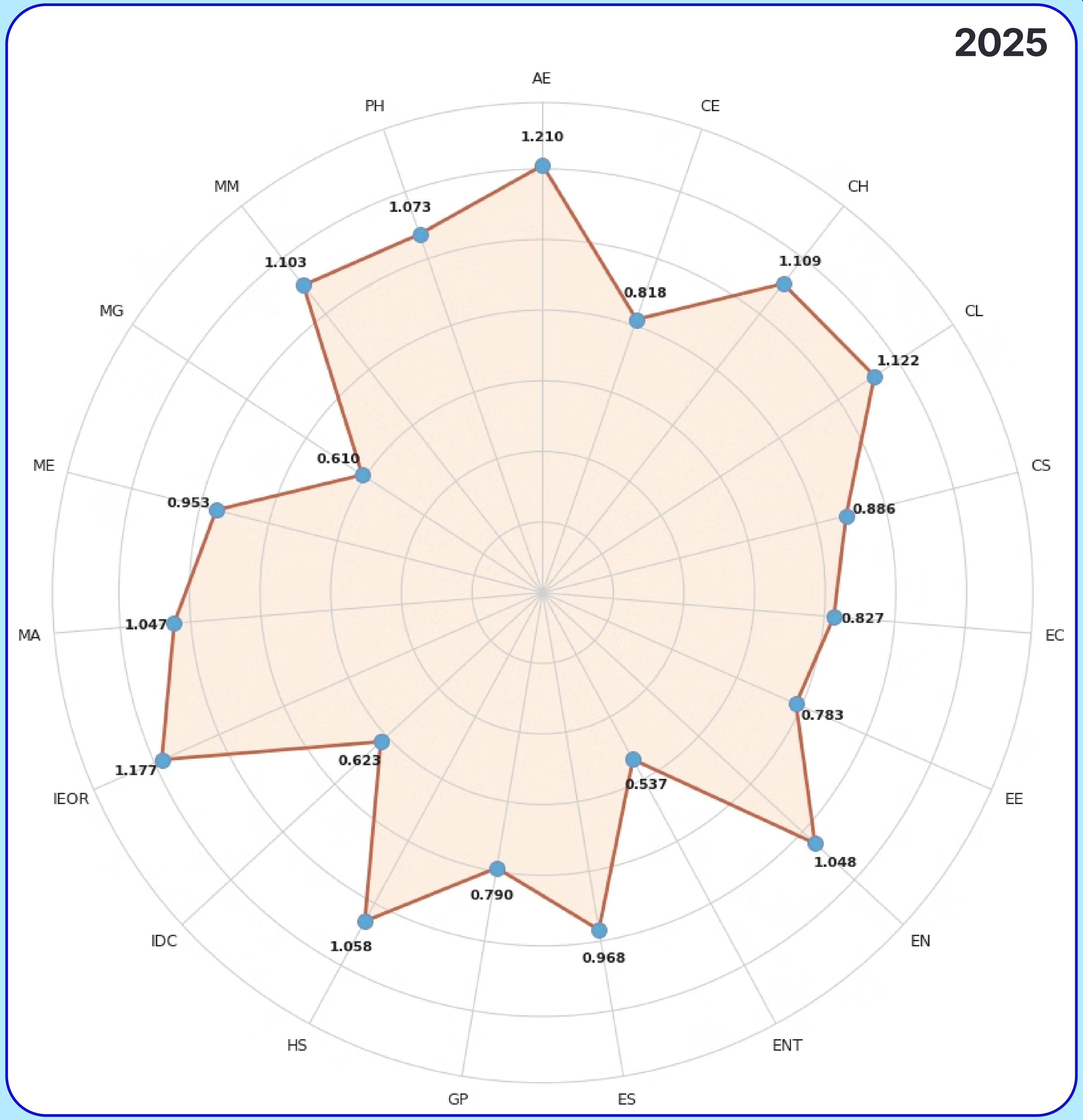
- ENT, IDC, and GP consistently tend to have high grades, with averages reliably exceeding the 8.0 mark.
- ES has the most consistent grades, with its four year average fluctuating by a mere 0.12 points. This highly predictable and stable pattern is closely matched by departments like EC, MG, and CS.
- **The 2024 Macro Level Grade Expansion:** The academic year 2024 marked a widespread upward shift in grades across the entire institute, with 13 out of the 19 tracked departments recording a higher average grade compared to their 2023 baselines.

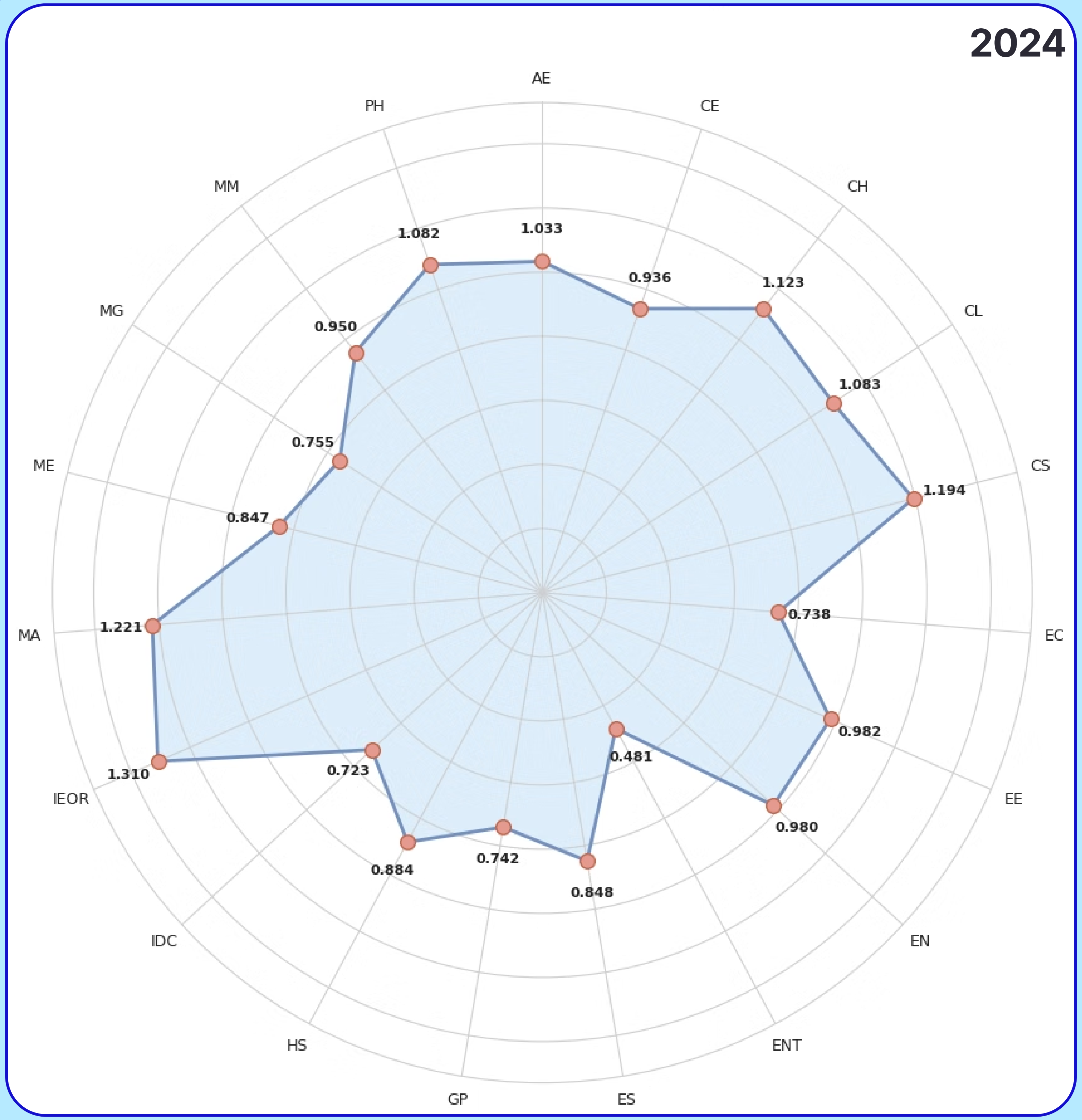
NOTE:

1. The department wise averages shown in the previous 2 graphs are calculated globally by aggregating all course types offered by each specific department, including mandatory core courses, departmental electives and minor courses.
2. The division of departments into "Group 1" and "Group 2" is purely administrative to balance the visual layout across two slides; these groupings do not reflect any underlying metric, category or academic classification.

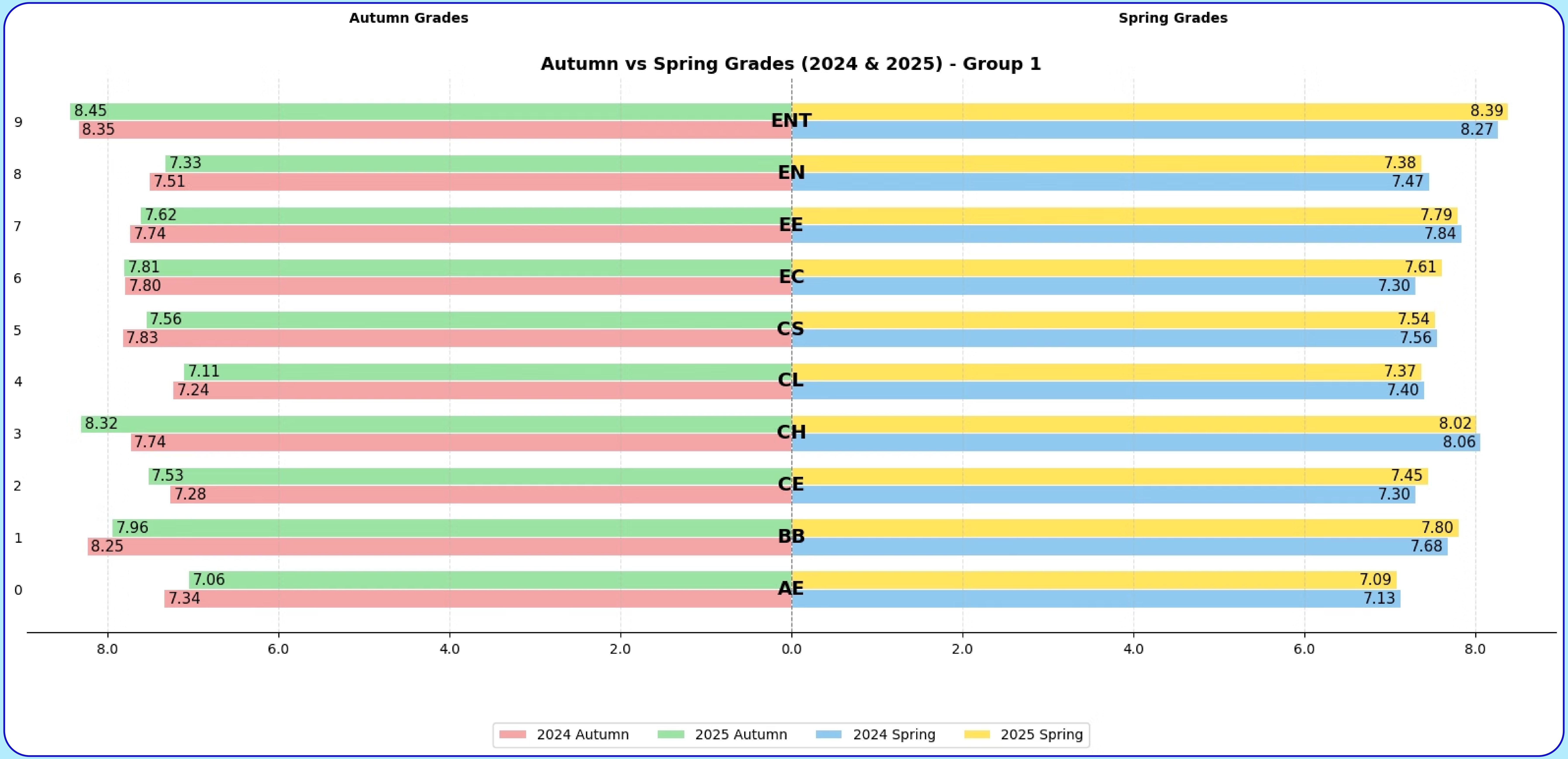
Department Wise Standard Deviation of Average Grades

Reflects the full spread of grading across every course the department has offered.

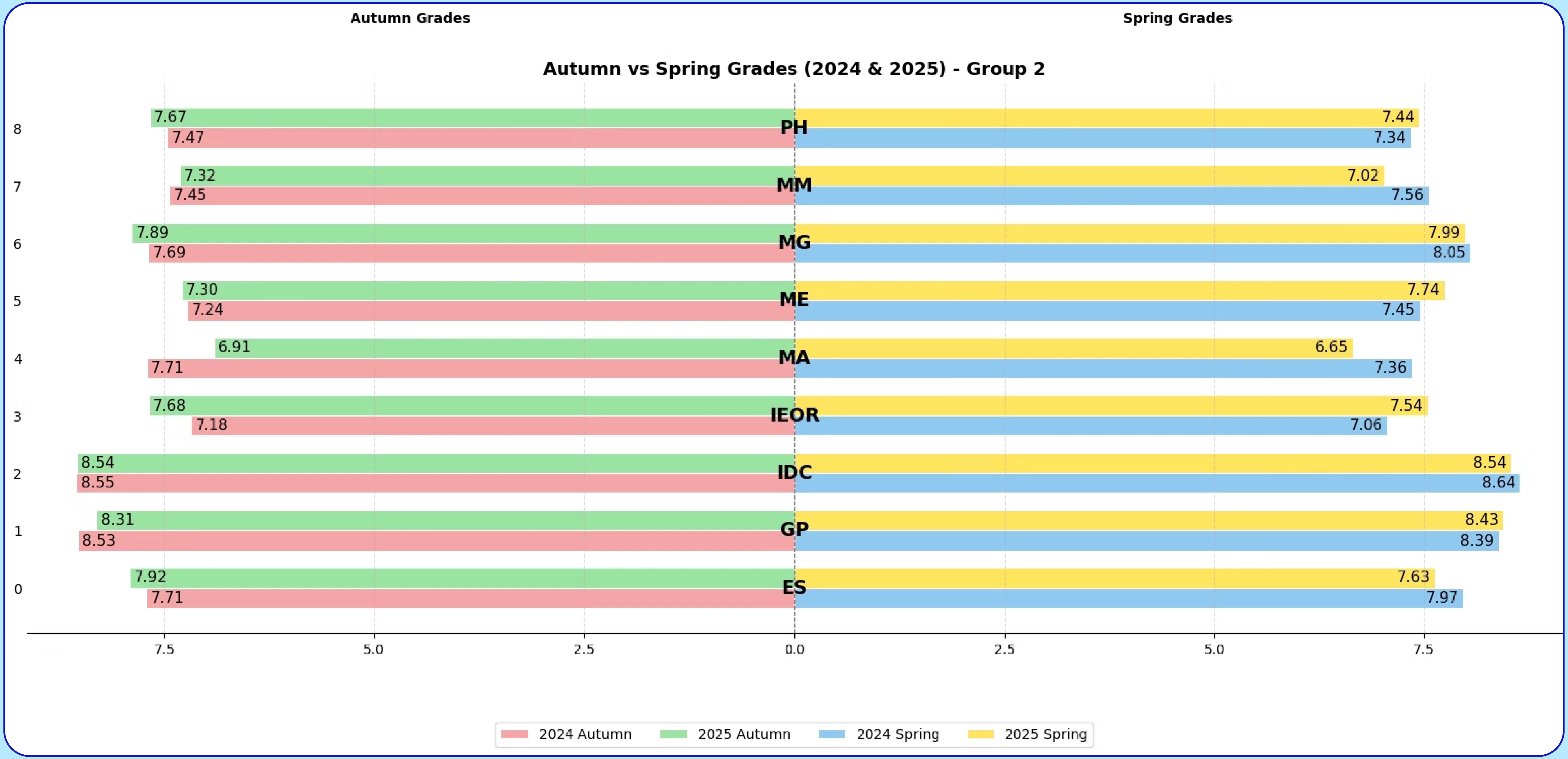




Average Grades of Department: Autumn vs Spring, 2024 vs 2025



Average Grades of Department: Autumn vs Spring, 2024 vs 2025



Department wise AP Grade Distribution

AP Grade Utilization by Department (2022-2025)

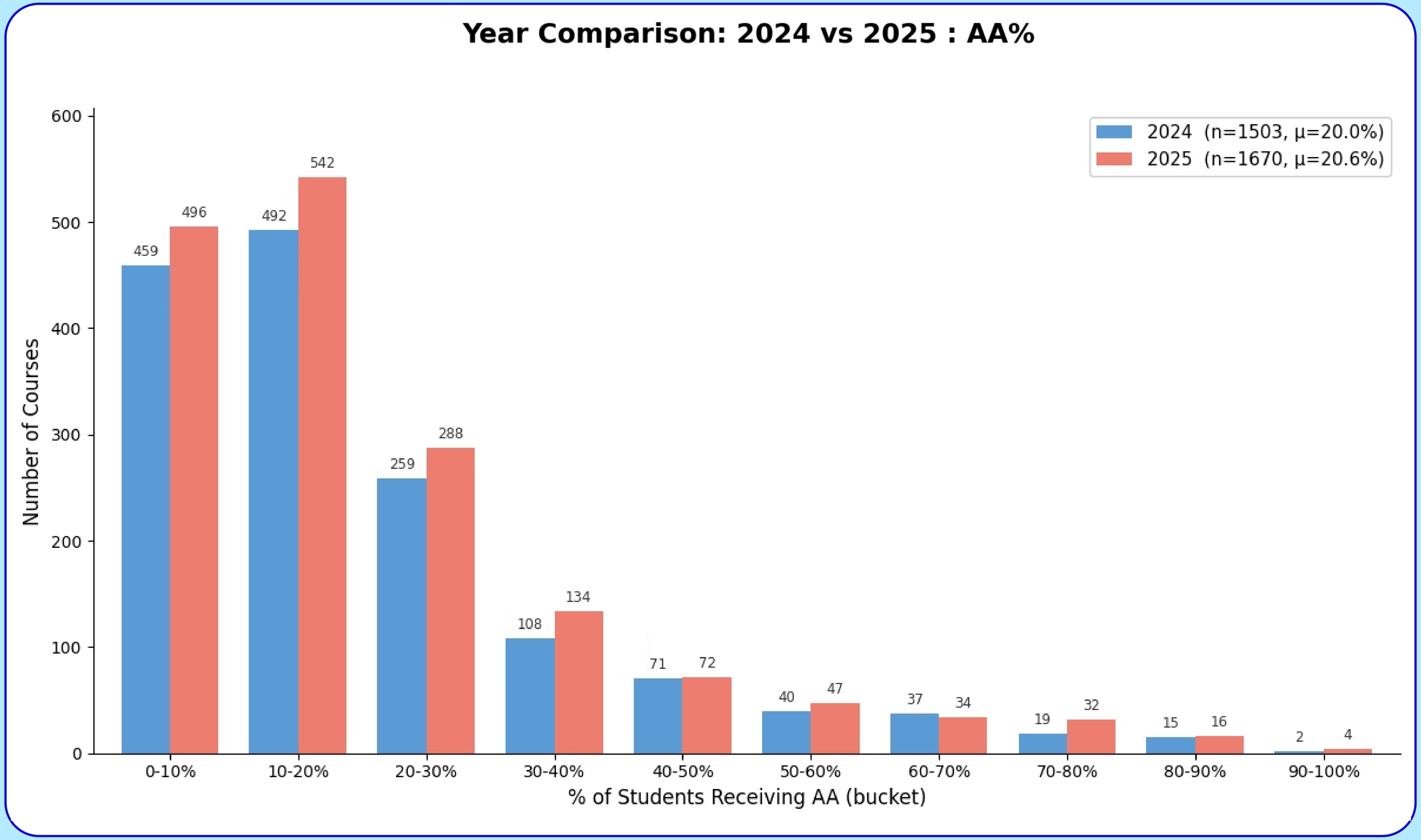
[AP Util % = AP Count / (2% × Students) × 100 | courses with >50 students only]

	2022	2023	2024	2025
AE	13.91	12.14	7.66	12.39
BB	50.30	34.13	68.63	47.27
CE	17.57	23.46	20.46	21.86
CH	45.56	23.92	22.55	27.23
CL	28.14	21.58	27.46	38.25
CS	40.04	37.64	45.91	35.02
EE	29.00	21.32	27.40	27.32
EN	0.00	7.63	0.00	10.94
ENT	50.44	72.57	76.02	69.08
ES	0.00	3.43	20.16	16.75
IDC	0.00	0.00	5.34	4.88
IEOR	9.88	26.39	13.85	13.79
MA	49.69	50.73	37.80	29.61
ME	24.82	23.79	16.44	12.77
MG	11.98	3.64	4.34	7.83
MM	17.70	19.82	23.89	33.58
PH	26.15	17.45	22.16	19.82

The maximum no. of APs that can be given in a course is 2% of registered students, thus the numbers in the cells represent the percentage of APs being given relative to total capacity of APs that can be given. Only the courses with more than 50 registrations are considered for this analysis.

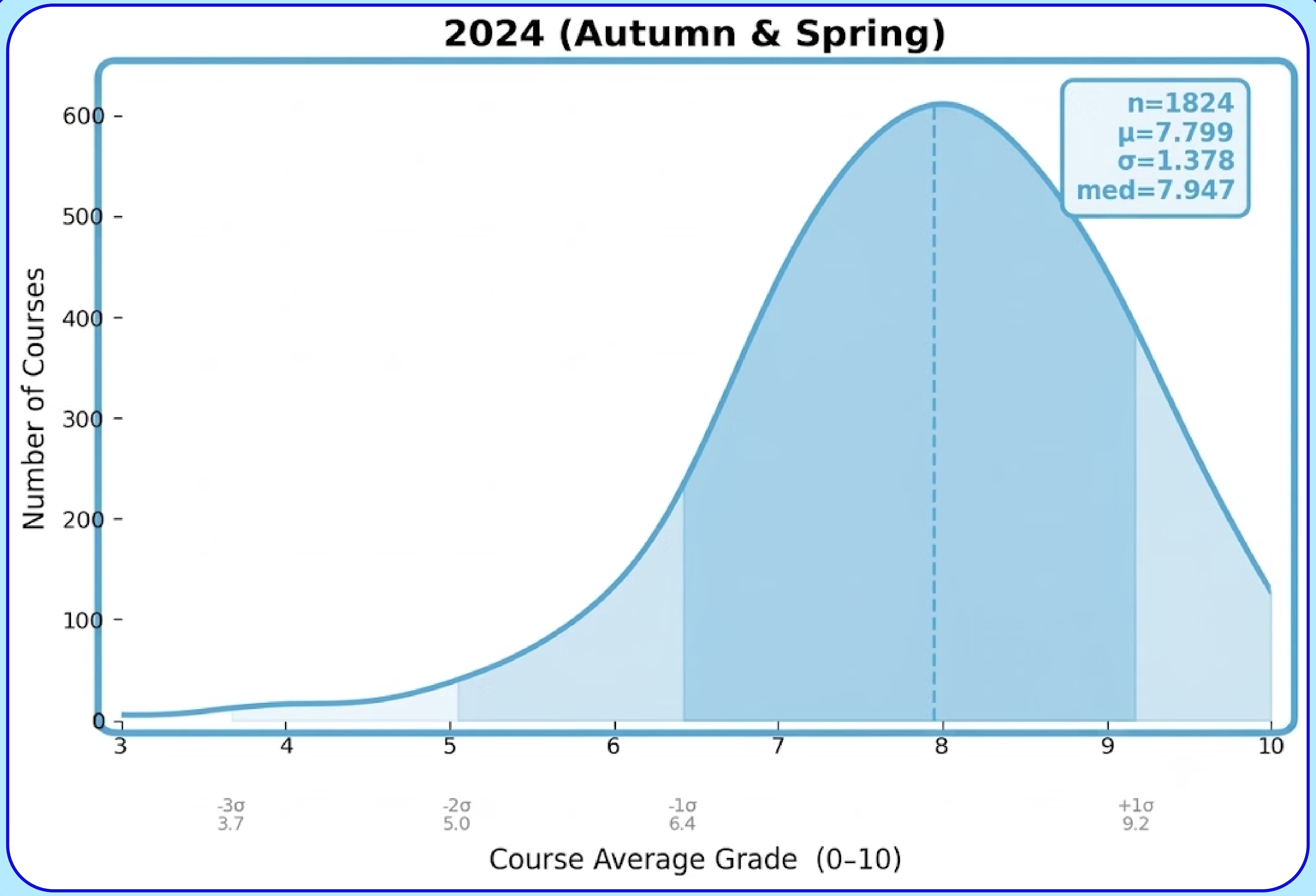
- ENT and BB consistently maximize their allocated AP grades.
- MA shows a clear and continuous multi year decline in awarding AP grades.
- EN, ES, and IDC courses award near zero AP grades.

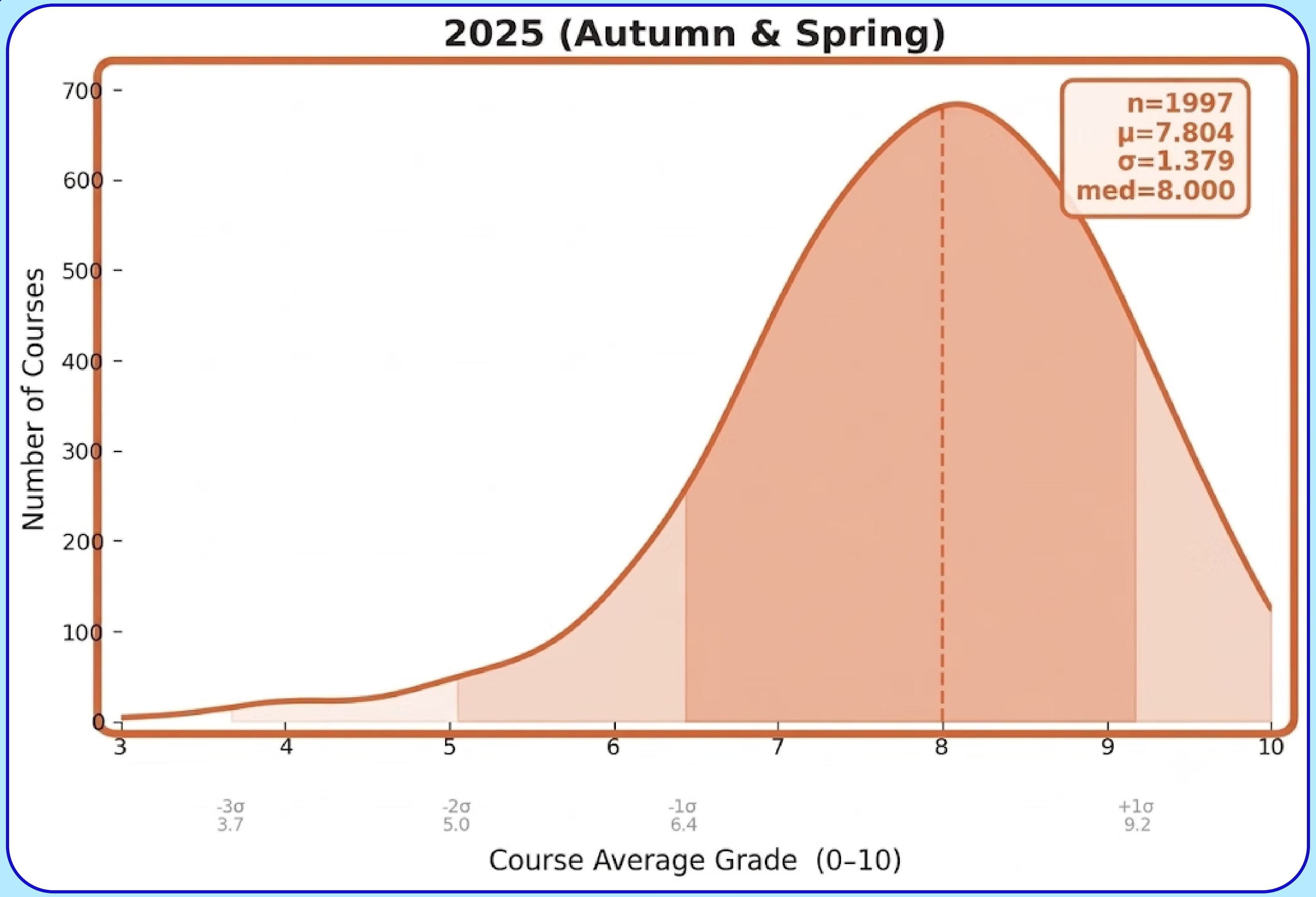
Number of Courses vs Percentage of Students Receiving AA



- Both years exhibit a steep right skewed distribution, demonstrating that the vast majority of courses award the top "AA" grade to only a small fraction of students; specifically, a massive chunk of offerings cluster tightly within the 0-20% bracket.
- In 2025, the 10–20% bin saw the most substantial growth.

Number of Courses vs Average Grades





- Course offerings increased by 173 in 2025 (n=1824 to 1997). During this period, the mean (7.799 to 7.804) and standard deviation (1.378 to 1.379) remained relatively stable.
- The median increased from 7.947 to 8.000, indicating that at least 50% of the courses in 2025 recorded an average of 8.0 or higher.

Course Recommender 3.0

To facilitate informed decision making when selecting Elective and Minor courses, students are advised to utilize the Course Recommender Tool, developed by the Data Analytics and Visualization (DAV) Team. The latest version 3.0 would be released shortly. Key features of 3.0 include:

- **Tailored Recommendations:** The tool suggests courses based on individual academic interests, allowing users to refine their results by filtering specific departments, course popularity, lenient grading and other parameters.
- **Minor Course Identification:** It clearly identifies and lists courses that are offered as a Minor(both M-tagged and electives).
- **Conflict Detection:** The system automatically highlights potential scheduling conflicts between prospective electives/ minors and mandatory core courses.
- **Prerequisite Information:** It outlines the necessary prerequisites required prior to enrolling in specific courses.

Coming Soon!!

Thank You